TECHNICAL SPECIFICATION • IEEE 830 STANDARD FORMAT

Software Requirements Specification (SRS)

MemoryVault AI — Context-Aware Misplaced Item Prediction & Heuristic Search Router

Project: MemoryVault AI	**Version:** 1.0 (MVP)	**Target Runtime:** Python 3.10+, Flask	**Baseline Corpus:** ~5,000 Synthetic Samples

1. Introduction & System Overview

MemoryVault AI is a predictive, conversational assistant that determines the probable indoor location of misplaced everyday items (such as keys, wallets, and earphones) and calculates an optimal, sequential physical search trajectory. Rather than relying on expensive hardware tags or manual logging, the system infers placement using behavioral context modeling, multiclass classification, and state-space graph search.

The architecture couples an unsupervised learning pipeline for habit pattern discovery, a multiclass Random Forest classifier for zone prediction, an A* / Greedy Best-First graph search router for inspection pathing, and a lightweight Flask REST API with a responsive web chat interface.

2. User Roles & Permissions

Role	Description	Permissions & Capabilities
End User	Individual seeking lost items within a mapped residential environment.	Submit natural language queries, answer clarifying context prompts, view ranked search routes, and record binary placement feedback.
System Evaluator / Admin	Technical administrator or researcher evaluating model performance.	Trigger training runs, retrieve confusion matrices, review cross-validation & ROC-AUC scores, and configure environment graphs.

3. Functional Requirements (FR)

Module 1: Conversational Chatbot Interface

- **FR-01 (Natural Language Query Intake):** The system shall parse unstructured user inputs (e.g., *"Where are my keys?"*) via a web chat interface.
- **FR-02 (Dynamic Contextual Probing):** The engine shall dynamically identify missing parameters and prompt the user with at most two clarifying questions (e.g., arrival time, weather, baggage carried).
- **FR-03 (Search Plan Rendering):** The client shall present an ordered, step-by-step physical inspection checklist with predicted zone probability percentages.

Module 2: Machine Learning & Behavioral Intelligence

- **FR-04 (Unsupervised Habit Clustering):** The system shall apply Principal Component Analysis (PCA) for dimensionality reduction and K-Means clustering to discover at least 3 distinct contextual habit archetypes.
- **FR-05 (Supervised Location Classification):** A multiclass Random Forest model shall map multidimensional context vectors to probability distributions over discrete domestic zones.
- **FR-06 (Model Evaluation Pipeline):** The system shall run 5-fold Stratified Cross-Validation, generating Confusion Matrices and macro-averaged One-vs-Rest ROC-AUC curves.

Module 3: Heuristic Search Routing

- **FR-07 (Spatial Graph Representation):** Residential layouts shall be modeled as a weighted graph $G = (V, E)$, where V denotes inspectable candidate zones and E represents non-negative traversal costs.
- **FR-08 (A* / Greedy Best-First Search Routing):** The pathfinding engine shall calculate the optimal sequence of locations by balancing edge traversal cost inversely against target probability.

4. Business Rules (BR)

- **BR-01 (Clarification Limit):** The conversational agent must never issue more than two clarifying diagnostic questions per inquiry to avoid interaction fatigue.
- **BR-02 (Cost Function Formulation):** Graph traversal cost between nodes *u* and *v* shall be dynamically weighted using:

  ```
  Cost(u, v) = Distance(u, v) / (P(v) + ε), where ε = 0.01
  ```

  This ensures physically proximate, high-probability locations are checked before distant or improbable rooms.
- **BR-03 (Candidate Pruning Threshold):** Candidate zones with predicted probability $P(v) < 0.05$ shall be pruned from physical routing calculations.
- **BR-04 (Cold-Start Fallback):** Users with fewer than 5 recorded placement interactions shall default entirely to synthetic corpus prior weights.

5. Data Requirements & Validation

Context vectors must conform to strict schema requirements before inference:

Feature Name	Type	Accepted Values & Constraints
`item_type`	String (Enum)	`keys`, `wallet`, `earphones`, `spectacles`, `backpack`
`time_of_day`	Float	0.00 – 23.99 (fractional 24-hour timestamp)
`day_of_week`	Integer	0 (Monday) through 6 (Sunday)
`weather_condition`	String (Enum)	`clear`, `rain`, `snow`, `extreme_heat`
`carrying_baggage`	Boolean	`true` if carrying groceries/packages; else `false`
`fatigue_level`	Integer	1 (Energetic) to 5 (Exhausted)

Dataset Requirements: The synthetic training corpus must contain ≥ 5,000 balanced records with no class representing > 30% or < 5% of samples.

6. Security, Authentication & Data Privacy

- **Stateless Sessions:** Ephemeral client sessions managed using UUID session tokens stored in browser memory.
- **Role-Based Access:** Non-authenticated users restricted to basic chat and search endpoints; evaluation metrics (ROC-AUC, confusion matrices) secured under `ROLE_ADMIN`.
- **Local Processing:** All household topological data and habits remain local to the Flask instance; no spatial telemetry is transmitted externally.

7. Performance & Non-Functional Requirements (NFR)

Identifier	Metric	Acceptance Threshold
NFR-01	End-to-End Latency	< 2.0 seconds from chat prompt submission to checklist rendering.
NFR-02	Top-3 Classification Accuracy	≥ 80.0% accuracy on a 20% holdout test dataset.
NFR-03	Classifier ROC-AUC	Macro-averaged One-vs-Rest ROC-AUC ≥ 0.85 across all zones.
NFR-04	Search Trajectory Efficiency	A* path distance shorter than random check orders in ≥ 95.0% of test cases.
NFR-05	Resource Consumption	Flask backend and model runtime memory footprint ≤ 2 GB RAM.

8. Error Handling & Edge Cases

- **Unrecognized Item:** Returns an HTTP 400 error indicating unsupported item types and lists available tracked items.
- **Disconnected Graph Component:** If target zones fall into unreachable nodes, A* issues a fallback to probability-only rank ordering.

- **High-Entropy Predictions:** When all candidate zones have $P(v) < 0.20$, the chatbot explicitly surfaces low confidence and advises checking primary landing zones.

9. Verification & Traceability Matrix

Req ID	Target Component	Verification Method
VR-01	Supervised Model	5-fold cross-validation confirming Top-3 Accuracy ≥ 80% and ROC-AUC ≥ 0.85.
VR-02	Habit Clustering	K-Means generating ≥ 3 distinct habit clusters confirmed via PCA 2D/3D scatter visualization.
VR-03	A* Search Pathing	Algorithmic evaluation verifying cumulative cost lower than baseline permutation in ≥ 95% of runs.
VR-04	Flask Chatbot Interface	Automated integration test verifying HTTP 200 payload return within 2.0 seconds.